MODULE 2 · LECTURE 11

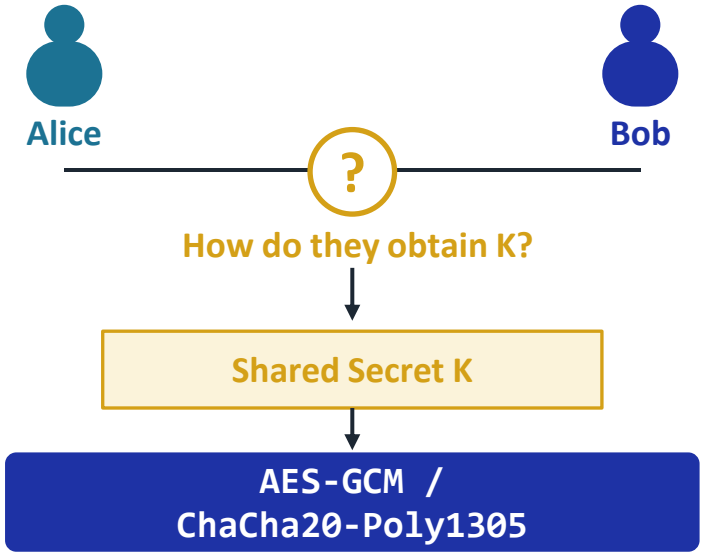
CS 6530 Applied Cryptography

Public-Key Cryptography: Concepts and Applications

Module 3 – Public-Key Cryptography

Why Public-Key Cryptography?

Symmetric cryptography efficiently protects data — once communicating parties share a secret key.



Public-key cryptography provides:

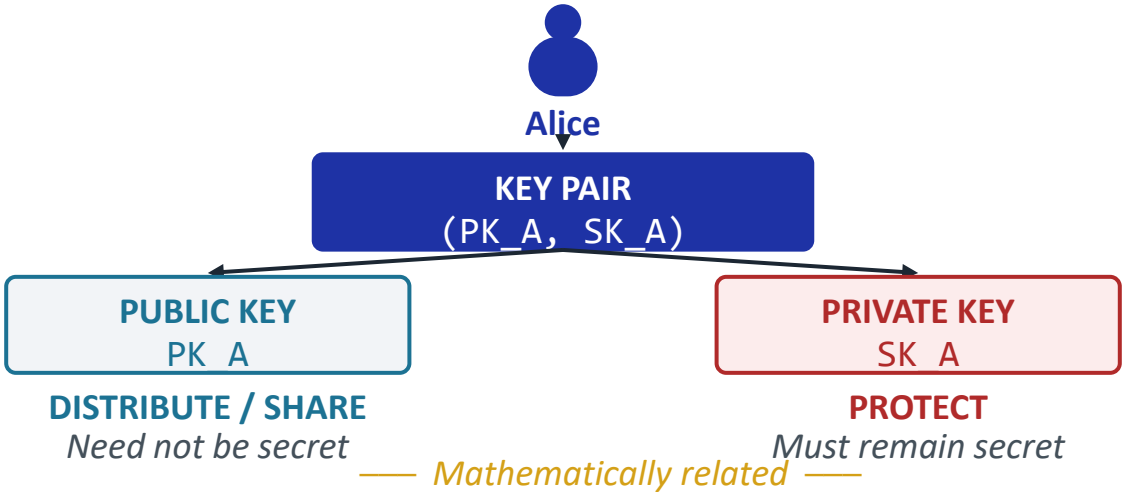
Key establishment · Encryption · Digital signatures

Think Like an Engineer

Two parties want to use AES-GCM, but have never communicated before. **What must happen first?**

Public-key cryptography complements symmetric cryptography; it does not replace it.

Public and Private Keys: The Asymmetric Idea



Given the public information, recovering the private key must be computationally infeasible.

Public ≠ authenticated
Having a public key does not by itself establish whose key it is.

Key authentication → Certificates / PKI later

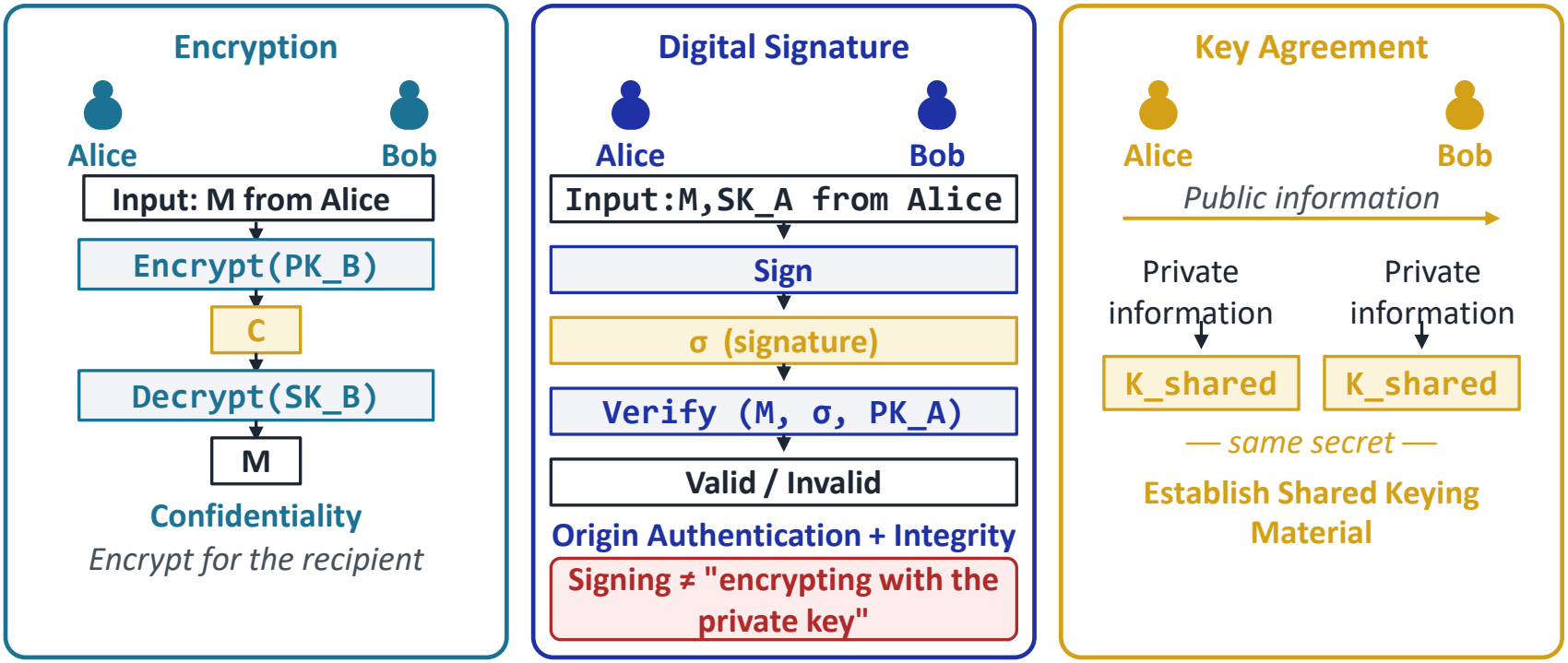
Think Like an Engineer

If everyone may know PK_A, what must the security of the system depend on?

Computational hardness + secrecy/protection of SK_A.

The public key is not secret. The private key is. Security relies on computational hardness.

Three Public-Key Capabilities



Different security objectives · Different constructions

Not every public-key algorithm provides every capability.

Think Like an Engineer

Which capability protects data for Bob? Which lets others verify Alice? Which helps Alice and Bob derive shared keying material?

Encrypt, sign, and agree on keys are distinct cryptographic capabilities — not interchangeable operations.

THE MATHEMATICAL IDEA

Computational Asymmetry: Easy One Way, Hard the Other



Forward computation is efficient



Recovering x from y is intended to be computationally infeasible.

One-Way-Function Intuition

Easy to compute in one direction; computationally hard to reverse under the intended security assumptions.

Some constructions add a trapdoor



Trapdoor: secret information that makes a hard operation feasible.

Not every public-key construction requires a trapdoor function.
RSA is a classical trapdoor-style construction; Diffie–Hellman-style key agreement exploits computational hardness differently.

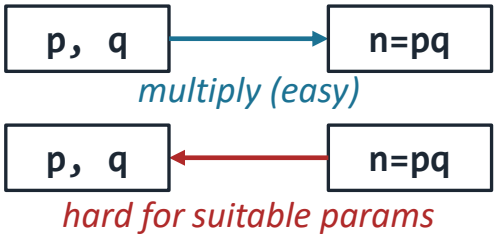
Think Like an Engineer

If everyone could efficiently reverse f using only public information, what happens to the security assumption?

Public-key cryptography exploits computational asymmetry; some constructions additionally use a secret trapdoor.

Different Constructions, Different Hard Problems

RSA



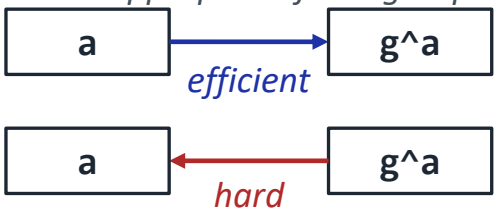
Hard-Problem Connection
Integer factorization

Construction Idea
Trapdoor-style structure

Mathematical foundations + RSA → next lecture

DIFFIE–HELLMAN (DH)

in an appropriate finite group



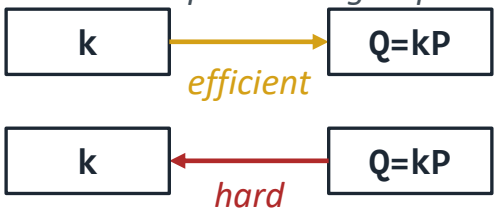
Hard-Problem Connection
Discrete logarithm

Construction Idea
Algebraic structure + computational hardness

Shared secrets derived without solving the DLP.

ELLIPTIC-CURVE CRYPTOGRAPHY (ECC)

in an elliptic-curve group



Hard-Problem Connection
Elliptic-curve discrete logarithm

Construction Idea
Group structure + computational hardness

Public-key constructions use different mathematical structures and hardness assumptions.
These classical assumptions weaken against capable quantum computers → Post-Quantum Cryptography later.

Think Like an Engineer

Does every public-key construction require an efficient way for the legitimate user to reverse its underlying hard problem?

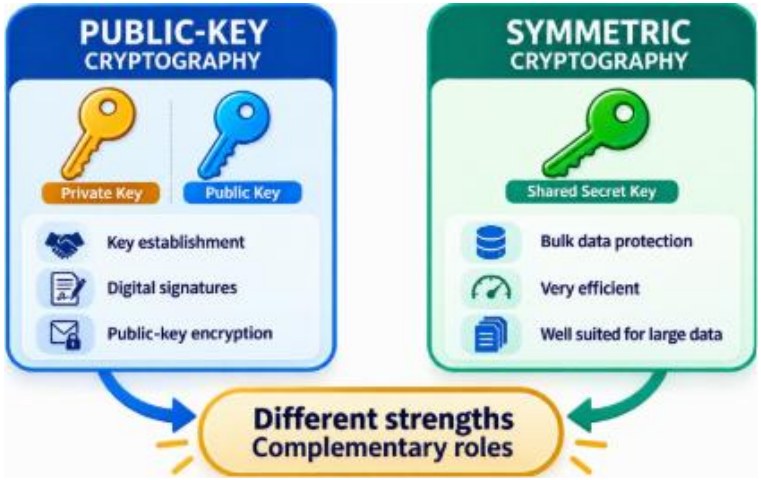
Computational hardness is the common theme; how each construction uses it is different.

THE ENGINEERING TRADE-OFF

Public-Key Cryptography Has a Cost

If public-key cryptography provides powerful capabilities, why not use it for all application data?

Symmetric Cryptography	Public-Key Cryptography
Shared secret	Public/private information
AES, ChaCha20	RSA, ECC-based constructions
Well suited to bulk data protection	Generally not used directly for bulk data
Very efficient	Relatively expensive
Does not provide digital signatures	Signature schemes provide digital signatures

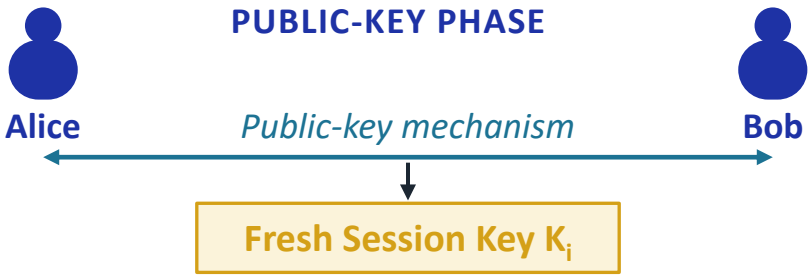


Think Like an Engineer

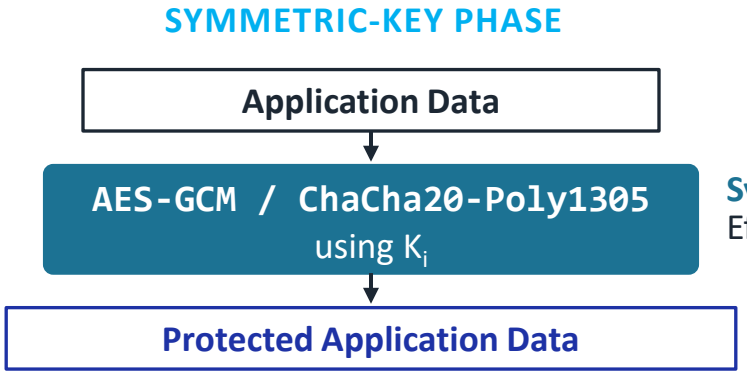
Would you encrypt a 2 GB file directly using RSA? What primitive would you choose instead?

Use public-key and symmetric cryptography for the roles each performs best.

Hybrid Cryptosystems: Combining the Best of Both



Public-key mechanisms : Establish or protect keying material. *Key establishment: e.g., key transport or key agreement*



Symmetric cryptography
Efficiently protects bulk application data.

Hybrid cryptography appears in secure communication systems such as: **TLS** • **SSH** • **Secure Messaging**

Think Like an Engineer

Suppose an attacker obtains the session key K_1 for one session. A later session independently establishes a fresh key K_2. Does compromise of K_1 reveal the communication protected using K_2? Why or why not?

Public-key mechanisms enable key establishment, encryption, and signatures; symmetric cryptography efficiently protects bulk data.